

Reconfigurable Switch: A Pure XDP Approach, Design and Implementation

Abstract

New technologies bring benefits to life and work as well as security threats and risks. Despite the increasing investment in Cyber security, security incidents occurred continuously. Lack of effective lateral movement identifying and controlling is a major cause. In this paper, by leveraging a pure XDP approach, we present the design of the reconfigurable switch which provides the abilities to program and orchestrate functional modules to implement not only functions like traditional switches but also access control from layer 2 through layer 7. We then demonstrate how to implement the design by building a prototype. We also provide a performance evaluation of the prototype showing that reconfigurable switch can be used as an infrastructure device with access control ability as well as an acceptable performance.

批注 [KC1]: Here, should we evaluate the performance or should we demonstrate our abilities to stop lateral movement inside local network?

Introduction

With new technologies emerging, such as AI, Big Data, IoT, and so on, our businesses are more efficient, and our life becomes more convenient. However, the concerns on cyber security come together with the benefits. The attacks on cyber security, such as APT or ransomware can damage data, privacy and secrets, assets, finances, and even more people's lives. Therefore, more and more cyber security devices or techniques are implemented in our production environments. Even the AI are introduced into cyber security domain to help people analyze the data and logs to identify the potential attacks or security breaches.

Major measures for cyber security work on board protection, which prevents attackers from outside. While others work on endpoint protection, which ensures the endpoint which connects to the network is secured. However, they are challenged. With more and more devices connected into the environment, the board is not as clear as old-time. Controlling a single entrance helps with very little on security. Endpoint protection is also restricted because it relies on agent working on the protected endpoints. However,

those agents may not be able to be installed onto IoT devices as well as BYOD. Therefore, new protection philosophies are raised up, e.g. Zero-Trust, which aims to eliminate the disadvantages of traditional cyber security. Zero-Trust brings a no-board theory, so the protection does not rely on the board check but requires verification of each request everywhere at any time. It does help esp. on roaming working scenarios. However, for most industrial environments and IoT environments, zero-trust is too complex to implement. For those organizations who have implemented Zero-Trust, only less than 30% have network microsegment enabled.

Despite so many cyber security investments, security incidents still occur on vary environments. Lateral movement is one of the main causes of those attacks. With so many devices (protected or unprotected) connected in the network, attackers can always find a way to break in, e.g. phishing, sending malicious software by emails, IM tools, or portable storage, or even connecting a malicious device into the network. Although the initial broken point is usually worthless, it can be used as an intermediate to attack other valuable assets. Malicious actors leverage the lateral movement approach to perform reconnaissance tasks to locate high-value devices or data, to widespread automation tools or bots to other vulnerable devices, and ultimately to perform the attack as desired. Research showed that over 70% of successful attacks leveraged lateral movement techniques. By examining the whole life cycle of an attack incident, it appears that malicious actors spend about 80% of the time doing lateral movement tasks. Barracuda reports that about 44% of ransomware attacks were identified during lateral movement. Therefore, it is essential for organizations to identify and prevent lateral movements to protect the environment from attacking. To do so, EDR/XDR relies on endpoint agents installed; internal firewalls need to segment network into different zones. Even with zero-trust, the situation is the same. It seems that we lack measure to identify and prevent lateral movement over the network. This is because we are still focusing on the boards and endpoints and require them to be the execute point of any security policies.

Switches which connect the whole network can be the great place as the policy executing point if the security mechanism of firewalls, EDR/XDR, and/or other security devices can be added up. It is necessary to have a switch which can not only process packet forwarding effectively and efficiently as usual but also perform security checks against the packets and execute access control policies. Programmable switches look like a doable approach while traditional programmable switches are not for common usage

批注 [KC2]: <https://www.elisity.com/blog/the-top-11-cyberattacks-using-lateral-movement-a-2023-2024-analysis-for-enterprise-security-leaders#:~:text=Lateral%20movement%20%20poses%20a,operational%20disruption%2C%20and%20compliance%20fines>

批注 [KC3R2]: This article will be referred further for listing lateral movement attacks.

due to cost and complexity. In this article, we introduce the reconfigurable switch, which is completely designed based on XDP (eXpress Data Path), an eBPF-based high-performance network data path to transfer network packets at high rates. The reconfigurable switch can work as usual switch, such as, to forward packets at layer 2 and layer 3. Besides that, it also provides the following abilities. First, it can implement packet examination and policy execution at layer 2 through layer 7 by attaching necessary function modules. Second, it provides the interfaces that administrators can leverage to develop their own function modules. Third, administrators can also orchestrate the attached function modules according to their own requirement to achieve high-speed packet processing and security goals as desired.

The rest sections are organized as follows. In Section 2, we provide an overview of XDP and related concepts. Section 3 presents the whole picture of the designation of the reconfigurable switch. We then demonstrate how the reconfigurable switch works by implementing a prototype in Section 4, following by a performance evaluation of the prototype in Section 5. Finally, Section 6 outlines the future research and summarizes the key findings in the current work.

Background

XDP

XDP (eXpress Data Path) is a high performance and programmable data path technology, which operates at the early stage of the network driver, so it can process network packets before they reach the Linux kernel stack. This significantly reduces processing overhead and latency. XDP is built on eBPF (Extended Berkeley Packet Filter), which is a sandboxed virtual machine in the Linux kernel. This environment enables programmers to filter, redirect, modify and process packets directly in an efficient and secure manner. Therefore, XDP is widely used in DDoS mitigation, network load balancing, traffic monitoring, and edge computing.

The XDP system consists of four main components, which are the XDP driver hook, the eBPF virtual machine, BPF maps, and eBPF verifier. The XDP driver hook is the entrance of the XDP program, which is executed packets arrive at the XDP-attached network interface card. Written in C language usually, XDP programs are compiled into BPF byte codes and executed inside the eBPF virtual machine. The XDP programs are

triggered to execute when a packet arrives. Because they always start with the initial state, the XDP programs use BPF maps to store and retrieve data in program context. BPF maps can be used as global configuration storage, and the information share between user space programs and kernel eBPF programs. Types of maps commonly used in XDP are Hash Maps, Array Maps, and Ring Buffers. The former two maps are similar and useful for maintaining and sharing program context; while the latter one, Ring Buffers, enables high-speed communication between XDP and user space programs. Since XDP operates directly in the kernel's memory space, any inappropriate operation could lead to kernel panic. The eBPF verifier prevents such risks by enforcing a strict static analysis on XDP programs before they are loaded into the kernel. If any potentially unsafe operation, e.g. memory out-of-bound access, infinite loops, unauthorized function calls, or unsafe pointer dereferences, is detected, the eBPF verifier will stop loading the XDP program.

In the early days, an eBPF program is limited to no more than 4096 instructions. Tail calls were introduced to link several programs to overcome the limitations. Although the limitation has been enlarged to a million instructions, tail calls are still helpful to break down complex logic into smaller, composable eBPF programs. For instance, a set of tail calls can be used in XDP as a pipeline process to achieve packet parsing, access control, and other operations.

XDP programs stop execution after returning predefined actions. These actions indicate how a packet should be processed as the table shows below.

<i>Return Code</i>	Action
<i>XDP_ABORTED</i>	Terminates execution due to an error
<i>XDP_DROP</i>	Drops the packet (e.g., DDoS filtering)
<i>XDP_PASS</i>	Allows packet to enter kernel stack
<i>XDP_TX</i>	Sends packet back through the same interface
<i>XDP_REDIRECT</i>	Redirects packet to another NIC or AF_XDP

Table 1. Predefined actions for XDP programs.

Programmable Switch

```
// todo:
// explain programmable switch vs cyber security, how it can help
// and why it is not a usual solution
```

Lateral Movement

// todo:

// explain how lateral movement affects cyber security.

Design

Implementation

Evaluation

Future Research and Conclusion